

Causal-Enhanced Graph Reinforcement Learning for Cross-Domain Topology Optimization

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Abstract—In the era of 6G endogenous intelligence, wireless knowledge graphs (WKGs) are key to autonomous network operation. However, constructing high-fidelity WKGs is challenging: expert-defined topologies fail to capture dynamic dependencies, while data-driven approaches are vulnerable to noise. To address this, we propose Causal-RL-GNN, an evaluation-optimization framework for cross-domain topology construction. A causal-aware search space reduction mechanism prunes spurious correlations via lagged analysis. An inductive graph neural network serves as a transferable quality evaluator, enabling robust zero-shot assessment across domains. Based on this evaluation, topology construction is formulated as an iterative optimization problem and solved using a GraphSAGE-based reinforcement learning agent with entropy regularization and reward shaping. Experiments on real-world wireless datasets show that our framework achieves a Pearson correlation coefficient of 0.90 and successfully recovers key structural patterns.

Index Terms—Wireless Knowledge Graph, Causal Inference, Graph Neural Networks, Reinforcement Learning, Topology Optimization.

I. INTRODUCTION

The transition towards 6G networks is driven by the paradigm of endogenous intelligence [1], envisioning a network capable of autonomous perception and self-evolution. To manage the complexity of ultra-dense deployments, wireless knowledge graphs (WKGs) have emerged as a critical technology for root cause analysis and resource scheduling [2], [12].

However, constructing high-fidelity WKGs and evaluating their reliability remain significant challenges [3]. Existing approaches mainly focus on graph construction. Current industry practices rely on expert-defined rules derived from static protocols (e.g., 3GPP). While accurate for physical adjacencies, they fail to capture dynamic and probabilistic dependencies such as interference coupling [4]. On the other hand, data-driven methods attempt to infer topology from Key Performance Indicator (KPI) time-series data, but often suffer from noise and spurious correlations. A quantitative mechanism is needed to assess whether the constructed topology aligns with the dynamic physical environment [13].

Inferring potential relationships from multiple time series can be viewed as a classic causal discovery problem.

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Causal discovery methods, such as Granger Causality [5] and constraint-based approaches like PC and PCMC [6], [9] either assume linearity or rely on strict conditional independence, making them sensitive to noise and less effective in complex wireless environments. In our task, we can adopt a coarse-to-fine strategy, first generating a search space with high recall. This serves as a constraint, ensuring that subsequent optimization operations remain within physically feasible bounds.

In graph generation, deep generative models like VGAE [7] and GraphRNN [14] excel at mimicking training data distributions but are unsuitable for optimizing sparse, incomplete backbones towards structural robustness. While RL methods like GCPN [15] allow for goal-directed generation, their success depends on the availability of a domain-specific reward function, which is absent in wireless networks.

To bridge this gap, we propose a novel framework, termed **Evaluation-Driven Graph Optimization (Causal-RL-GNN)**. Unlike traditional generation methods, we formulate topology construction as an iterative refinement process. Starting from a sparse expert-defined backbone, we first establish a transferable evaluation standard and then employ a Deep Reinforcement Learning (DRL) agent [8] to sequentially optimize the structure, maximizing the assessed quality under causal constraints.

Our approach uniquely integrates three complementary paradigms to address WKG topology optimization:

1. **Causal-Aware Pruning**: Instead of computationally expensive full-graph search, we leverage lagged correlation analysis to reduce the search space. Unlike precision-oriented methods such as PCMC [9], we prioritize recall to construct a high-coverage candidate set while filtering global trend noise.
2. **Inductive Evaluation**: To address the lack of ground truth in new deployment scenarios, we employ a Graph Isomorphism Network (GIN) [10] as a transferable critic. By incorporating topological invariants (e.g., PageRank), the model enables robust zero-shot evaluation across domains.
3. **Generative Optimization**: Guided by the evaluator, a GraphSAGE [11]-based agent iteratively refines the topology. We adopt the REINFORCE algorithm with entropy regularization to effectively explore the discrete combinatorial space, balancing structural sparsity and global connectivity.

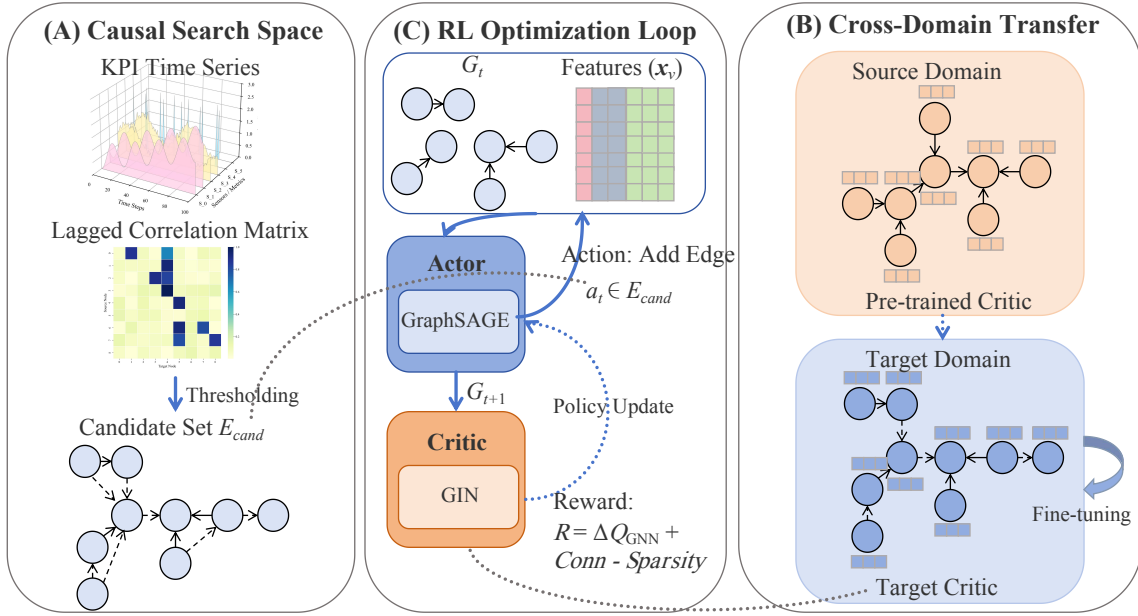


Fig. 1: Architecture of the Proposed Causal-RL-GNN.

II. METHODOLOGY

We propose a closed-loop framework for Cross-Domain Network Topology Optimization, termed **Causal-RL-GNN**. To address the challenge of missing ground-truth labels in target domains, we adopt an evaluation-driven paradigm. As illustrated in Fig. 1, the framework operates through three coupled stages:

- (A) **The Constraint:** A causal search space module that mines time-series data to prune irrelevant connections, ensuring physical plausibility;
- (B) **The Critic:** An inductive graph quality assessment model that serves as a transferable proxy evaluator, establishing a universal standard for structural robustness;
- (C) **The Actor:** A generative topology optimization module via robust DRL that iteratively refines the network structure under the guidance of the Critic.

A. Problem Formulation

Let $G_{base} = (V, E_{fixed})$ denote the expert-defined backbone network, where V is the set of N network entities, and E_{fixed} represents the immutable physical connections. Each node $v_i \in V$ is associated with a multivariate KPI time-series $\mathbf{X}_i = [x_{i,1}, \dots, x_{i,T}] \in \mathbb{R}^T$, capturing the dynamic behaviors of the entity.

Our objective is to identify an optimal set of missing edges E_{opt} from the vast combinatorial space of potential links. The reconstructed graph $G^* = (V, E_{fixed} \cup E_{opt})$ must maximize structural robustness and functional efficiency while satisfying physical constraints. It can be expressed as a constrained maximization problem:

$$\max_{E_{opt}} \mathcal{J}(G^*) \quad \text{s.t.} \quad \begin{cases} G^* \text{ is globally connected,} \\ |E_{opt}| \leq K, \end{cases} \quad (1)$$

where $\mathcal{J}(\cdot)$ is a comprehensive quality function. Since the ground-truth topology for the target domain is unknown, we approximate $\mathcal{J}$ using a learned neural network and solve the discrete optimization problem using Reinforcement Learning.

B. Causal-Aware Search Space Constraint (The Constraint)

Blindly searching for edges in the full combinatorial space of $\mathcal{O}(N^2)$ is computationally intractable for RL and prone to overfitting noise. To constrain the optimization landscape, we employ a data-driven pruning mechanism to generate a high-recall candidate set E_{cand} . This acts as a physical validity mask for the RL agent.

1) *First-Order Differencing:* KPI data in wireless networks often exhibits strong non-stationarity and global co-movements. These common drivers lead to spurious correlations where disconnected nodes appear statistically related. To eliminate these confounding global trends and isolate intrinsic fluctuations, we apply first-order differencing:

$$\Delta \mathbf{X}_i(t) = \mathbf{X}_i(t) - \mathbf{X}_i(t-1), \quad (2)$$

where $\mathbf{X}_i(t)$ represents the raw metric value of node i at time step t , and $\Delta \mathbf{X}_i(t)$ represents the instantaneous rate of change.

2) *Robust Lagged Correlation Analysis:* Interactions between network entities are not always instantaneous; they often manifest with time delays. Unlike strict Granger causality, we adopt a robust lagged correlation metric. We compute the maximum absolute Pearson correlation coefficient between the differenced series of node pairs over a permissible time lag window $\tau \in [1, \tau_{max}]$:

$$S_{i \rightarrow j} = \max_{\tau} |\text{Corr}(\Delta \mathbf{X}_i(t - \tau), \Delta \mathbf{X}_j(t))|, \quad (3)$$

where $S_{i \rightarrow j}$ represents the causal confidence score indicating the strength of the potential link from node i to j .

3) *Candidate Generation*: A dynamic thresholding strategy retains the top- K percentile of edges to form the candidate set $E_{cand} = \{(i, j) \mid S_{i \rightarrow j} > \delta\}$. This ensures the ground-truth topology is encompassed within the search space while reducing dimensionality from $\mathcal{O}(N^2)$ to $\mathcal{O}(|E_{cand}|)$.

C. Inductive Graph Quality Assessment (The Critic)

To guide the optimization process in the target domain where ground-truth topologies are unavailable, we propose a **transferable Inductive GNN**. Unlike transductive models that memorize node identities, this Critic learns to evaluate structural robustness based on topological invariants and statistical distributions, enabling zero-shot transfer from the source domain.

1) *Topological Feature Augmentation*: Standard GNNs struggle to distinguish between structural noise and functional connections solely from raw time-series embeddings. To bridge this gap, we design a composite feature vector $\mathbf{z}_v$ for each node v , fusing statistical signal embeddings with explicit topological invariants:

$$\mathbf{z}_v = \text{Concat}(\mathbf{h}_v^{\text{stat}}, \log(\deg(v) + 1), C_{\text{clust}}(v), P_{\text{rank}}(v), H_{\text{homo}}(v)). \quad (4)$$

The components are defined as follows:

- **Statistical Embeddings ($\mathbf{h}_v^{\text{stat}}$)**: Z-score normalized statistics (Mean, Std, Max, Min) of the KPI time-series $\mathbf{X}_v$.
- **Log-Degree ($\log(\deg(v) + 1)$)**: Captures the fundamental connectivity density. The logarithmic scaling is applied to handle the power-law distribution typical in scale-free networks.
- **Local Clustering Coefficient (C_{clust})**: Measures the cliquishness of a local neighborhood, defined as:

$$C_{\text{clust}}(v) = \frac{2|\{e_{jk} : v_j, v_k \in \mathcal{N}(v), e_{jk} \in E\}|}{\deg(v)(\deg(v) - 1)}. \quad (5)$$

- **PageRank Centrality (P_{rank})**: Quantifies the global importance of a node based on the recursive definition:

$$P_{\text{rank}}(v) = \alpha \sum_{u \in \mathcal{N}_{\text{in}}(v)} \frac{P_{\text{rank}}(u)}{\deg_{\text{out}}(u)} + \frac{1 - \alpha}{N}, \quad (6)$$

where α is the damping factor (set to 0.85).

- **Feature Homophily (H_{homo})**: We define homophily as the average cosine similarity between a node's statistical features and its neighbors':

$$H_{\text{homo}}(v) = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} \frac{\mathbf{h}_v^{\text{stat}} \cdot \mathbf{h}_u^{\text{stat}}}{\|\mathbf{h}_v^{\text{stat}}\| \|\mathbf{h}_u^{\text{stat}}\|}. \quad (7)$$

Spurious edges added by random perturbations often link statistically dissimilar nodes, resulting in low H_{homo} values.

2) *Discriminative Architecture*: We employ a GIN backbone due to its superior discriminative power for graph structures [10]. Let $\mathbf{h}_v^{(0)} = \mathbf{z}_v$ denote the initial node representation derived from the augmented features. To address the internal covariate shift caused by the distinct traffic patterns between Uplink and Downlink scenarios, we replace the standard batch normalization with layer normalization (LN).

The node update rule at layer k ($k = 1, \dots, K$) is defined as:

$$\mathbf{h}_v^{(k)} = \text{LN}\left(\text{MLP}^{(k)}\left((1 + \epsilon^{(k)}) \cdot \mathbf{h}_v^{(k-1)} + \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(k-1)}\right)\right). \quad (8)$$

To capture both the average quality and salient structural defects (e.g., disconnected components), we utilize a readout function that concatenates global mean and max pooling:

$$\mathbf{h}_G = \text{Concat}\left(\frac{1}{|V|} \sum_{v \in V} \mathbf{h}_v^{(K)}, \max_{v \in V} \mathbf{h}_v^{(K)}\right). \quad (9)$$

Finally, the quality score is predicted via an MLP regressor:

$$Q(G) = \sigma(\text{MLP}(\mathbf{h}_G)) \in [0, 1]. \quad (10)$$

The quality label y is a weighted fusion of structural metrics of randomly perturbed graphs:

$$y = w_1 \cdot \text{F1} + w_2 \cdot \text{Jaccard} + w_3 \cdot \text{Precision} + w_4 \cdot \text{Recall}. \quad (11)$$

3) *Unsupervised Domain Adaptation*: We utilize the target domain's backbone as a pseudo-anchor to fine-tune the Critic. This aligns the model with the target feature distribution without requiring ground-truth labels.

D. Generative Optimization (The Actor)

We formulate the topology construction as a sequential decision-making process modeled by a Markov Decision Process (MDP). To solve this, we employ a **Robust GraphSAGE-based Policy Gradient** framework. Driven by the dynamic feedback from the Inductive Critic, the agent iteratively optimizes the graph structure.

1) *MDP Formulation*: The optimization process is defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$:

- **State (s_t)**: The state at time step t consists of the current graph snapshot $G_t = (V, E_{\text{fixed}} \cup E_t)$ and the dynamically updated node feature matrix $\mathbf{Z}_t \in \mathbb{R}^{N \times d}$.
- **Action (a_t)**: The action is to select a discrete edge e_{uv} to add to the graph from the available search space. The action space is constrained by the causal candidate set E_{cand} and the current edges E_t :

$$\mathcal{A}_t = \{(u, v) \mid (u, v) \in E_{cand} \setminus E_t\}. \quad (12)$$

- **Policy (π_θ)**: The probability of selecting an edge e_{uv} is modeled by a MLP taking the concatenated embeddings

of node pairs. To ensure validity, we apply a masked softmax:

$$\pi_\theta(a_t = e_{uv} | s_t) = \frac{\exp(\psi_{uv}^{(t)}) \cdot \mathcal{M}_{t,uv}}{\sum_{(i,j) \in E_{cand}} \exp(\psi_{ij}^{(t)}) \cdot \mathcal{M}_{t,ij}},$$

$$\psi_{ij}^{(t)} = \text{MLP}(\mathbf{h}_i^{(t)} \parallel \mathbf{h}_j^{(t)}). \quad (13)$$

where $\mathbf{h}_u^{(t)}$ is the embedding of node u at step t , and $\mathcal{M}_t$ is a binary mask vector where $\mathcal{M}_{t,ij} = 1$ if $(i, j) \in \mathcal{A}_t$ and 0 otherwise.

2) *Generative Architecture*: The Actor network (π_θ) consists of two components: a GraphSAGE Encoder for state representation and an MLP Decoder for action scoring.

We utilize GraphSAGE, an inductive embedding framework, to serve as the state encoder. For a node v , the embedding generation at layer l follows the message-passing paradigm:

$$\mathbf{h}_{\mathcal{N}(v)}^{(l)} = \text{AGGREGATE}^{(l)} \left(\left\{ \mathbf{h}_u^{(l-1)}, \forall u \in \mathcal{N}(v) \right\} \right),$$

$$\mathbf{h}_v^{(l)} = \sigma \left(\mathbf{W}^{(l)} \cdot \text{Concat} \left(\mathbf{h}_v^{(l-1)}, \mathbf{h}_{\mathcal{N}(v)}^{(l)} \right) \right), \quad (14)$$

where $\mathcal{N}_t(v)$ denotes the neighbors of v in the current graph G_t , and $\mathbf{W}^{(l)}$ are trainable weight matrices. We stack $L = 2$ layers to capture the 2-hop structural context. The final output $\mathbf{H}_t = \{\mathbf{h}_v^{(L)}\}$ serves as the context-aware state representation.

3) *Reward Shaping*: We design a composite reward r_t that balances structural quality, physical connectivity, and efficiency:

$$r_t = \alpha \cdot \Delta Q(G_t) + \beta \cdot R_{conn}(G_t) + \eta \cdot C_{uv} - \delta \cdot \rho(G_t) \quad (15)$$

The components are defined as follows:

- **Quality Gain (ΔQ)**: The improvement in the structural quality score predicted by the pre-trained Critic Q_ϕ .
- **Connectivity Bonus (R_{conn})**: A heuristic bonus based on the Largest Connected Component (LCC), encouraging actions that bridge isolated sub-networks.
- **Causal Bias (C_{uv})**: This is the causal confidence score $S_{u \rightarrow v}$ derived in Stage A. It acts as a prior, biasing exploration towards links with strong data-driven evidence.
- **Sparsity Penalty (ρ)**: To prevent the generation of overly dense structures, we impose a quadratic penalty on graph density.

4) *Robust Optimization Strategy*: The objective is to maximize the expected cumulative reward with entropy regularization. Let $R_t(\tau) = \sum_{k=t}^T \gamma^{k-t} r_k$ denote the discounted return at step t for a trajectory τ . The objective function is defined as:

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [R_0(\tau)] + \lambda_{ent} \mathbb{E}_{s_t \sim \tau} [H(\pi_\theta(\cdot | s_t))], \quad (16)$$

where $H(\cdot)$ is the Shannon entropy of the policy distribution, and λ_{ent} controls the exploration strength.

We optimize θ using REINFORCE with Monte Carlo returns, and the entropy term encourages exploration to avoid premature convergence.

Finally, the parameters are updated at each training epoch k with a learning rate η_{lr} :

$$\theta_{k+1} \leftarrow \theta_k + \eta_{lr} \cdot \nabla_\theta J(\theta_k). \quad (17)$$

Furthermore, an exponential learning rate decay scheduler is applied. A higher learning rate is used in early episodes to explore global connectivity patterns, while a decayed rate is used in later stages to fine-tune local structural details, ensuring convergence to a stable and high-quality topology.

III. EXPERIMENTS

To comprehensively evaluate the efficacy and robustness of the proposed Causal-RL-GNN framework, we conducted extensive experiments on real-world 5G network datasets.

A. Experimental Setup

1) *Real-world Datasets*: The experiments utilized KPI data collected from an operational 5G Radio Access Network (RAN). The dataset is divided into two distinct domains to simulate the cross-domain transfer scenario:

- **Source Domain (Uplink, UL)**: Used for pre-training the GNN Critic. It contains $N_{src} = 49$ network entities with a complete set of known physical links (E_{src}). The time-series data $\mathbf{X}_{src}$ consists of uplink-specific metrics (e.g., UL Throughput, UL Interference Power).
- **Target Domain (Downlink, DL)**: The target for topology optimization. It contains $N_{tgt} = 57$ entities, a sparse, expert-defined backbone graph G_{base} (containing only 40 edges, highly fragmented) and multivariate downlink time-series $\mathbf{X}_{tgt}$.

2) Implementation Details:

- **Causal Discovery**: The max lag is set to $\tau = 1$, and the top 30% of correlated pairs are retained as the candidate set E_{cand} .
- **GNN Critic**: A 3-layer GIN with layer normalization is employed. The model is pre-trained on UL data and fine-tuned on the DL backbone for 20 epochs ($lr = 5 \times 10^{-5}$).
- **RL Agent**: The GraphSAGE-based actor ($lr = 0.005$) is trained for 100 episodes using REINFORCE. We set the entropy regularization coefficient $\lambda_{ent} = 0.01$ and reward weights $\alpha = 1.0, \beta = 0.5, \eta = 0.1, \delta = 0.05$.

B. Search Space Validation

Before deploying the RL agent, it is crucial to verify that the generated candidate set E_{cand} effectively covers the latent ground-truth connections. We compare our Lagged Correlation approach against two established baselines: Lasso Granger [16], a linear method using sparse regression, and PCMCi, a state-of-the-art constraint-based method designed to eliminate spurious correlations.

We evaluate the performance using receiver operating characteristic (ROC) curves and confidence heatmaps.

1) *ROC Analysis*: As shown in Fig. 3, our Lagged Correlation method achieves the highest area under the curve (AUC = 0.795), significantly outperforming PCMCI (AUC = 0.628) and Lasso Granger (AUC = 0.558). A critical observation lies in the high-recall region: at a False Positive Rate (FPR) of 0.2, our method achieves a Recall of approximately 0.65, whereas PCMCI and Lasso Granger stagnate below 0.4. This indicates that our method can retrieve the majority of true edges with a manageable level of noise.

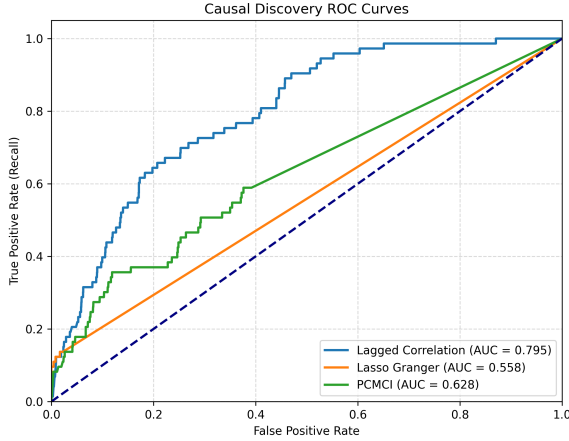


Fig. 3: ROC curves for causal discovery methods.

2) *Heatmap Visualization*: As shown in Fig. 2, the adjacency heatmaps provide a visual explanation for the quantitative results.

- **Ground Truth**: Exhibits a distinct block-diagonal structure with sparse local connections.
- **Lasso Granger**: Produces an extremely sparse matrix, missing most of the structural patterns.
- **PCMCI**: While capturing some diagonal elements, the signal is weak and blurred, failing to delineate the clear block structures.
- **Lagged Correlation (Ours)**: Successfully highlights the key block-diagonal regions (deep red areas) that align well with the Ground Truth. Although it introduces some

background noise (orange areas), it ensures that the true topology is encompassed within the search space.

C. Evaluation of Inductive Quality Assessor

A high-fidelity Critic is the cornerstone of our generative framework. We evaluate the GNN’s capability to predict structural quality scores under varying degrees of perturbation.

1) *Baseline Performance on Source Domain*: As illustrated in Fig. 4(a), the model achieves a Pearson Correlation Coefficient (PCC) of 0.9645 on the Source Domain. The predictions tightly align with the ideal $y = x$ diagonal, confirming that our feature augmentation strategy enables the GNN to capture the intrinsic semantics of structural robustness rather than memorizing node identities.

2) *Cross-Domain Generalization & Adaptation*: The challenge of domain shift is evident in the zero-shot transfer to the Target Domain. As shown in Fig. 4(b), the model without adaptation exhibits a systematic negative bias (predictions consistently below the diagonal), resulting in a high MSE of 0.0417. This bias suggests that while the model recognizes structural patterns (PCC = 0.7561 indicates preserved ranking ability), the distributional shift in KPI features causes the GNN to misinterpret valid structures as less reliable than they actually are.

By applying unsupervised fine-tuning on the target backbone, we effectively realign the feature space. As shown in Fig. 4(c), the bias is rectified, reducing the MSE by over 50% (to 0.0191) and boosting the PCC to 0.8856. The predictions regress strictly to the diagonal, demonstrating successful domain alignment.

3) *Significance for RL Optimization*: These improvements provide a more reliable reward signal for the RL agent, avoiding vanishing rewards under domain shift.

The improved PCC ensures that the Critic correctly identifies which topological changes lead to quality improvements, providing the correct gradient direction for the Actor. The reduced MSE and corrected bias prevent the “vanishing reward” problem, where an overly pessimistic Critic might discourage the agent from exploring valid connections. Thus, the adapted GNN serves as a robust evaluator, enabling the agent to navigate the combinatorial search space effectively even without ground-truth supervision.

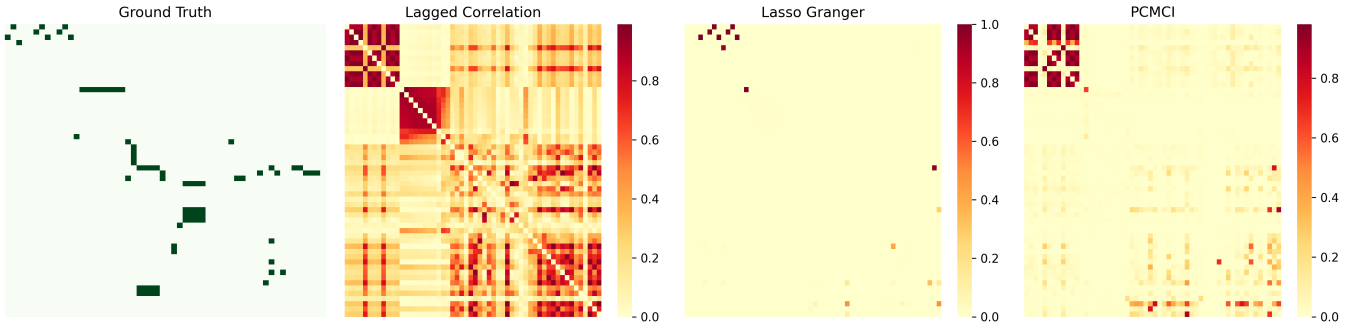


Fig. 2: Comparison of causal discovery heatmaps.



Fig. 4: Performance evaluation of the Inductive GNN Critic across domains.

D. Generative Topology Optimization

We analyze the performance of the GraphSAGE-RL agent in reconstructing the network backbone. The task is to transform a fragmented initial topology (G_{base} , 40 edges) into a fully connected, robust graph (G^*) using the causal candidate set (E_{cand}) as the search space.

1) *Optimization Dynamics*: The learning trajectory can be summarized as follows:

- **Connectivity Healing (Ep. 0–20)**: Driven by the connectivity bonus, the agent prioritizes “bridge edges”, rapidly boosting the connectivity ratio from 0.60 to near 1.0. This confirms the effectiveness of reward shaping in solving the fragmentation problem.
- **Structural Refinement (Ep. 20–60)**: Once connected, the agent enters an exploration phase (indicated by reward fluctuations), iteratively adding and pruning edges to maximize the GNN quality score.
- **Convergence (Ep. 60+)**: Aided by learning rate decay, the policy stabilizes, converging to a high-quality topology that balances structural robustness with sparsity constraints.

2) *Topological Evolution*: Table I visualizes the reconstruction process. The initial backbone (a) is sparse and disconnected. The intermediate state (b) achieves connectivity but contains redundant links. The final optimized topology (c) successfully prunes noise, recovering a clean block-diagonal structure that closely mirrors the Ground Truth (d).

Notably, the optimized graph recovers over 80% of the missing physical links. A small fraction of “mismatched” edges are transitive shortcuts (e.g., $A \rightarrow C$ instead of $A \rightarrow B \rightarrow C$), reflecting strong functional dependencies captured from data.

3) *Comparative Analysis with Baselines*: To validate the superiority of our framework, we compare it against three baselines: GCN-LP (Supervised Link Prediction) [17], [18], VGAE (Generative Model), and RL (No Transfer). Table II summarizes the quantitative results.

- **Global Connectivity**: Both GCN-LP and VGAE fail to ensure global connectivity (0.63 and 0.74, respectively). These models focus on local similarity or distribution matching, neglecting long-range bridges required to connect isolated sub-networks. Only RL-based methods achieve 100% connectivity.
- **Structural Fidelity (The Role of Transfer)**: While the ablation baseline RL (No Transfer) achieves connectivity, it does so by indiscriminately adding edges (Precision: 0.46), resulting in a noisy structure. In contrast, our proposed Causal-RL-GNN, guided by the transferred Inductive Critic, achieves a high F1-Score of 0.7975. This confirms that the Inductive Critic acts as a crucial Quality Gatekeeper, constraining the agent to select only physically meaningful connections.

In summary, our RL agent, stabilized by entropy regularization and connectivity-aware reward shaping, effectively balances the trade-off between sparsity and connectivity. It successfully constructs a cohesive network that exhibits high topological fidelity to the ground truth, surpassing the limitations of both stochastic and heuristic search methods.

IV. CONCLUSION

This paper proposes Causal-RL-GNN, a framework that refines static protocol-based skeletons into high-fidelity wireless knowledge graphs. By integrating recall-oriented causal pruning, we effectively reduce the combinatorial search space in high-noise environments. Furthermore, our proposed Inductive GNN Critic enables robust zero-shot transfer from source to target domains, solving the label scarcity challenge. Finally, the GraphSAGE-RL agent, stabilized by entropy regularization and reward shaping, successfully reconstructs physically valid and globally connected topologies. Future work will focus on capturing time-varying dependencies via dynamic graphs and integrating Large Language Models for semantic enhancement.

TABLE I: Evolutionary Process of Network Topology Optimization

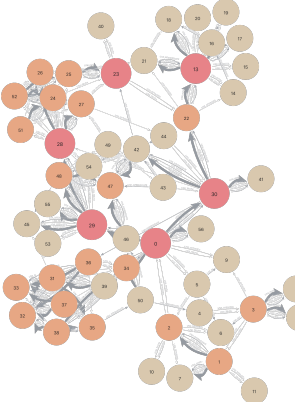
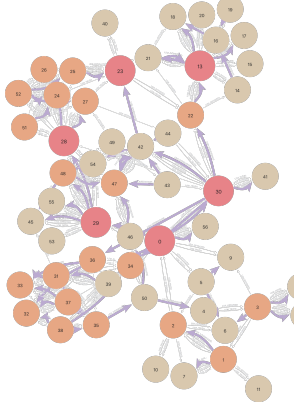
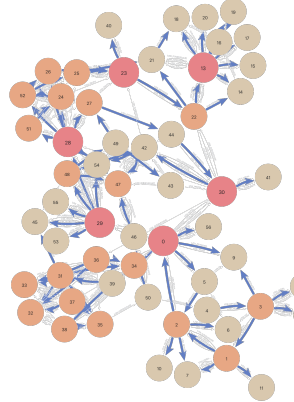
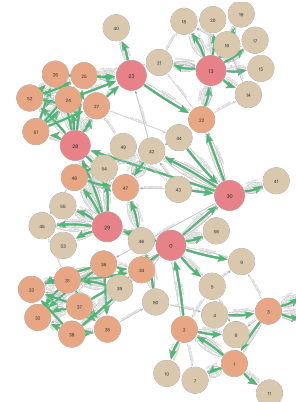
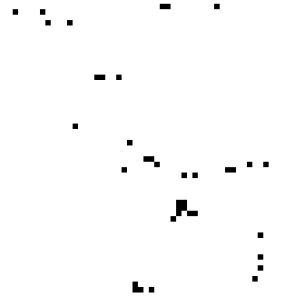
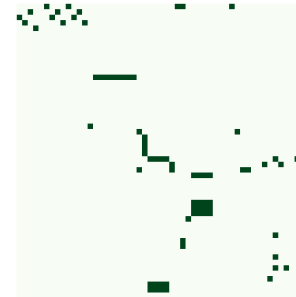
(a) Initial Backbone	(b) Intermediate State	(c) RL Optimized	(d) Ground Truth
			
			
Edges: 40 Conn.: 0.60 Q-Score: 0.35	Edges: 57 Conn.: 0.81 Q-Score: 0.62	Edges: 78 Conn.: 1.00 Q-Score: 0.88	Edges: 80 Conn.: 1.00 Q-Score: 1.00 (Ref)

TABLE II: Quantitative Comparison of Topology Reconstruction Methods

Method	Edges	Precision	Recall	F1-Score	Conn.
Initial Backbone	40	1.0000	0.5000	0.6667	0.5965
GCN LinkPred	80	0.5000	0.5000	0.5000	0.6316
VGAE Gen	78	0.5513	0.5375	0.5443	0.7368
RL (No Transfer)	89	0.4607	0.5125	0.4852	1.0000
Causal-RL-GNN	78	0.8077	0.7875	0.7975	1.0000

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